



ECEN 314 - Signals and Systems

Scott L. Miller

Module 13 - Frequency Domain Analysis of Circuits/Systems I
Transfer Functions, Magnitude/Phase Responses, Impulse Response

Learning Objectives



- Be able to find the transfer function associated with a circuit.
- Be able to find the transfer function associated with a system described by a differential equation.
- Given the transfer function of a system, be able to find the magnitude and phase response of the system.
- Be able to convert from magnitudes to decibels and from decibels to magnitudes.
- When using decibels, know when to use $10 \cdot \log$ and when to use $20 \cdot \log$.
- Given the transfer function of a system, be able to find the impulse response.

Frequency Domain Analysis



Consider a generic circuit/system with an input, $x(t)$, and an output, $y(t)$, whose relationship is described by a differential equation

$$\sum_{m=0}^M b_m \frac{d^m y}{dt^m} = \sum_{n=0}^N a_n \frac{d^n x}{dt^n}.$$

Take Fourier transforms of both sides of the equation. To simplify notation, define the Fourier transform operator, $\mathcal{F}[\]$, and the inverse Fourier transform operator, $\mathcal{F}^{-1}[\]$, as

$$\mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt,$$

$$\mathcal{F}^{-1}[X(f)] = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df.$$

In the frequency domain, the differential equation becomes

$$\mathcal{F}\left[\sum_{m=0}^M b_m \frac{d^m y}{dt^m}\right] = \mathcal{F}\left[\sum_{n=0}^N a_n \frac{d^n x}{dt^n}\right].$$

Next use various properties of Fourier Transforms to simplify.

Frequency Domain Analysis



$$\mathcal{F} \left[\sum_{m=0}^M b_m \frac{d^m y}{dt^m} \right] = \mathcal{F} \left[\sum_{n=0}^N a_n \frac{d^n x}{dt^n} \right].$$

Using linearity:

$$\sum_{m=0}^M b_m \mathcal{F} \left[\frac{d^m y}{dt^m} \right] = \sum_{n=0}^N a_n \mathcal{F} \left[\frac{d^n x}{dt^n} \right].$$

Using differentiation property:

$$\sum_{m=0}^M b_m (j2\pi f)^m Y(f) = \sum_{n=0}^N a_n (j2\pi f)^n X(f).$$

The output of the system (in the frequency domain) is then found according to:

$$Y(f) = \frac{\sum_{n=0}^N a_n (j2\pi f)^n}{\sum_{m=0}^M b_m (j2\pi f)^m} X(f).$$

Define the transfer function as:

$$H(f) = \frac{\sum_{n=0}^N a_n (j2\pi f)^n}{\sum_{m=0}^M b_m (j2\pi f)^m}$$

so that

$$Y(f) = H(f) X(f).$$

Frequency Domain Analysis



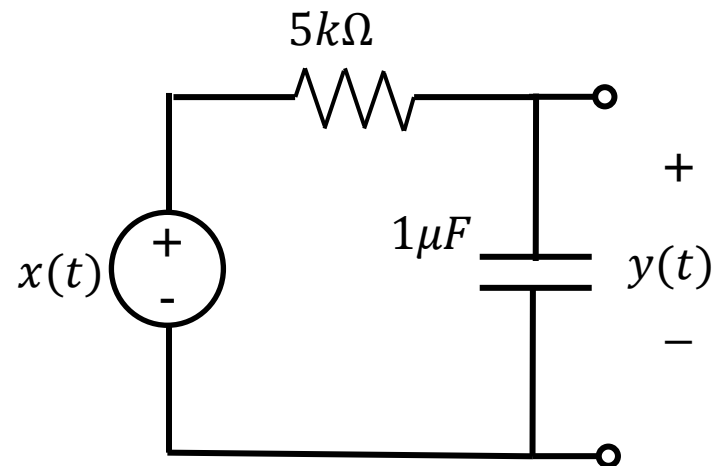
To find the output of the circuit/system due to the input $x(t)$, we can use a frequency domain approach using these steps:

- ❑ Step 1: Find the Fourier transform of the input signal - $X(f) = \mathcal{F}[x(t)]$,
- ❑ Step 2: Obtain the Fourier transform of the output using $Y(f) = H(f) X(f)$
- ❑ Step 3: Obtain the output in the time domain using an inverse Fourier transform:
 $y(t) = \mathcal{F}^{-1}[Y(f)] = \mathcal{F}^{-1}[H(f) X(f)]$
- ✓ Using frequency domain tools such as Fourier transforms, Laplace transforms, etc allow us to transform a differential equation into an algebraic one. This can greatly simplify the math as long as the transforms in steps 1 and 3 are manageable.
- ✓ Using the Fourier transform instead of the Laplace Transform has some advantages and disadvantages:
 - ❖ Using Fourier transforms, the forward and inverse transforms are essentially the same. Inverse Laplace transforms are quite a different animal from Laplace transforms.
 - ❖ Fourier transforms have a more physical interpretation (frequency content).
 - ❖ Fourier/Laplace transforms can each handle some signals the other cannot.

Frequency Domain Analysis



Example 13.1: Use Fourier transforms to find the output for the first order RC circuit shown if the input is $x(t) = e^{-100|t|}$



Answer: $y(t) = \left(2e^{-100t} - \frac{4}{3}e^{-200t}\right)u(t) + \frac{2}{3}e^{100t}u(-t)$

Transfer Functions

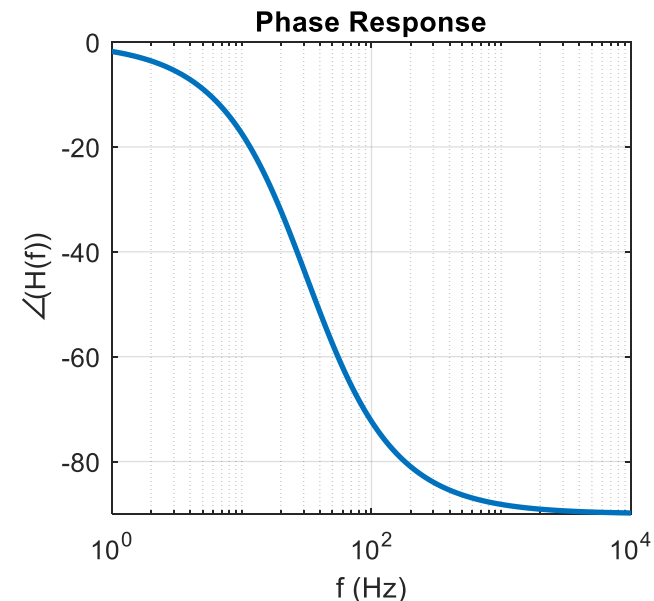
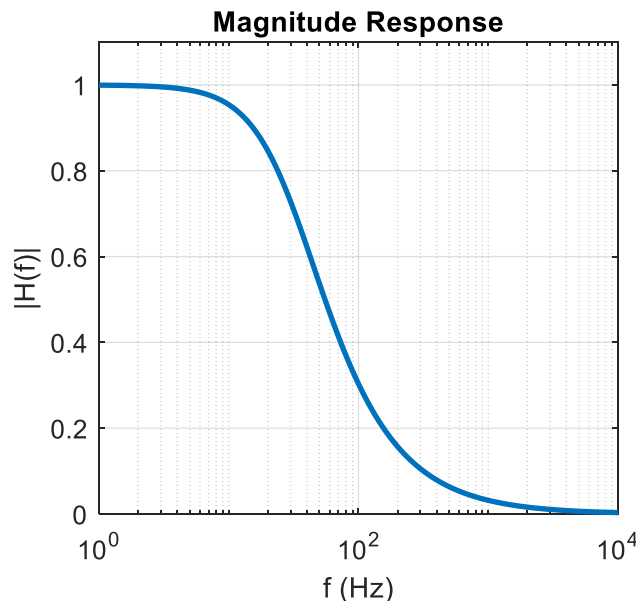


For the previous example, the transfer function of the RC circuit was

$$H(\omega) = \frac{200}{200+j\omega} \text{ or equivalently } H(f) = \frac{200}{200+j2\pi f}.$$

In general, the transfer function is a complex function of frequency. It is common to decompose this into two real functions in polar form:

- ✓ Magnitude response: $|H(\omega)| = \frac{200}{\sqrt{(200)^2 + \omega^2}}$ or $|H(f)| = \frac{200}{\sqrt{(200)^2 + (2\pi f)^2}}$
- ✓ Phase response: $\angle H(\omega) = -\tan^{-1}\left(\frac{\omega}{200}\right)$ or $\angle H(f) = -\tan^{-1}\left(\frac{\pi f}{100}\right)$



Magnitude Response - LPF



For the previous example, $H(\omega) = \frac{200}{200+j\omega}$:

- ✓ For low frequencies ($\omega \ll 200$), $|H(\omega)| \approx 1 \rightarrow |Y(\omega)| \approx |X(\omega)|$,
- ✓ For high frequencies ($\omega \gg 200$), $|H(\omega)| \approx 0 \rightarrow |Y(\omega)| \approx 0$.

This circuit functions as a low-pass filter. The low-frequency content of the input passes through and appears at the output, while the high-frequency content is blocked and does not appear at the output.

Suppose the positions of the resistor and the capacitor were exchanged. Then

$$H(\omega) = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega}{\frac{1}{RC} + j\omega} = \frac{j\omega}{200 + j\omega} \rightarrow |H(\omega)| = \frac{\omega}{\sqrt{(200)^2 + \omega^2}}.$$

In this case,

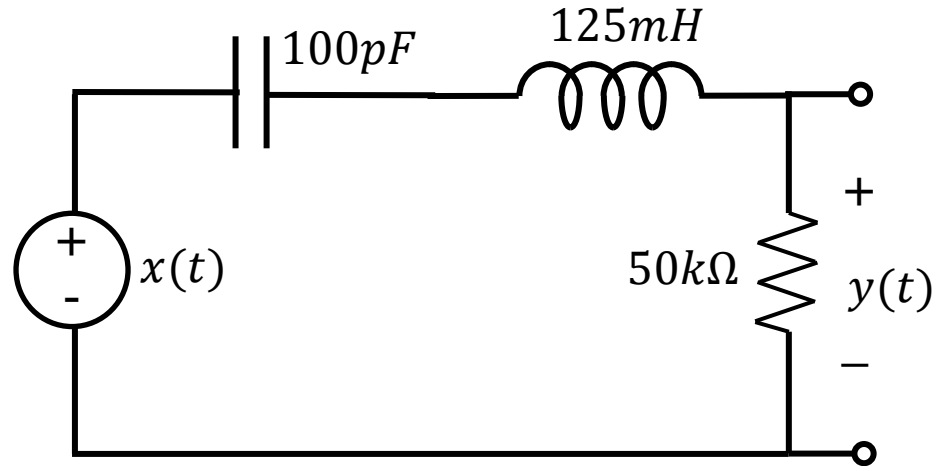
- ✓ For low frequencies ($\omega \ll 200$), $|H(\omega)| \approx 0 \rightarrow |Y(\omega)| \approx 0$
- ✓ For high frequencies ($\omega \gg 200$), $|H(\omega)| \approx 1 \rightarrow |Y(\omega)| \approx |X(\omega)|$.

Now the circuit functions as a high-pass filter. The low frequency content of the input is blocked while the high frequency content appears at the output.

Transfer Functions



Example 13.2: Find the transfer function, magnitude response, and the phase response of the RLC circuit shown.



Answers:

$$H(\omega) = \frac{j\omega}{200,000 - \frac{\omega^2}{400,000} + j\omega},$$

$$|H(\omega)| = \frac{\omega}{\sqrt{\left(200,000 - \frac{\omega^2}{400,000}\right)^2 + \omega^2}},$$

$$\angle H(\omega) = \frac{\pi}{2} - \tan^{-1} \left(\frac{\omega}{200,000 - \frac{\omega^2}{400,000}} \right).$$

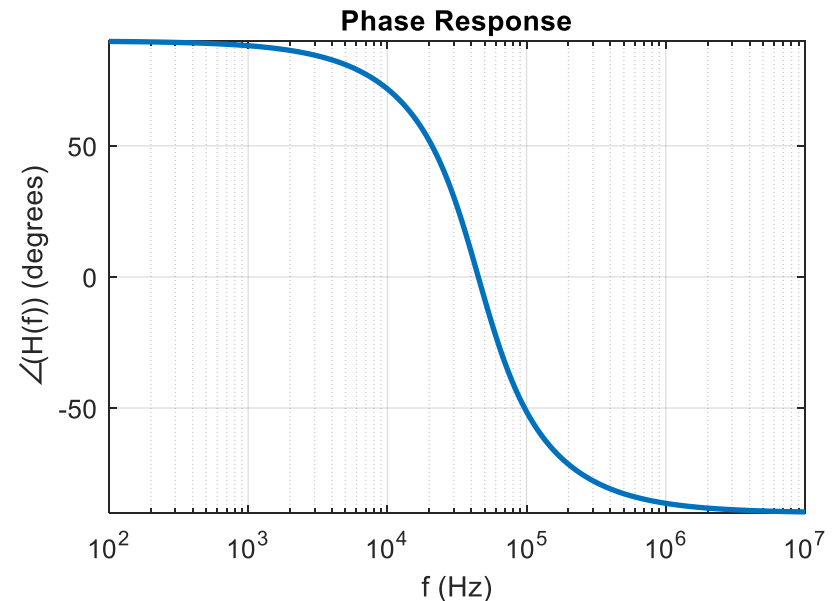
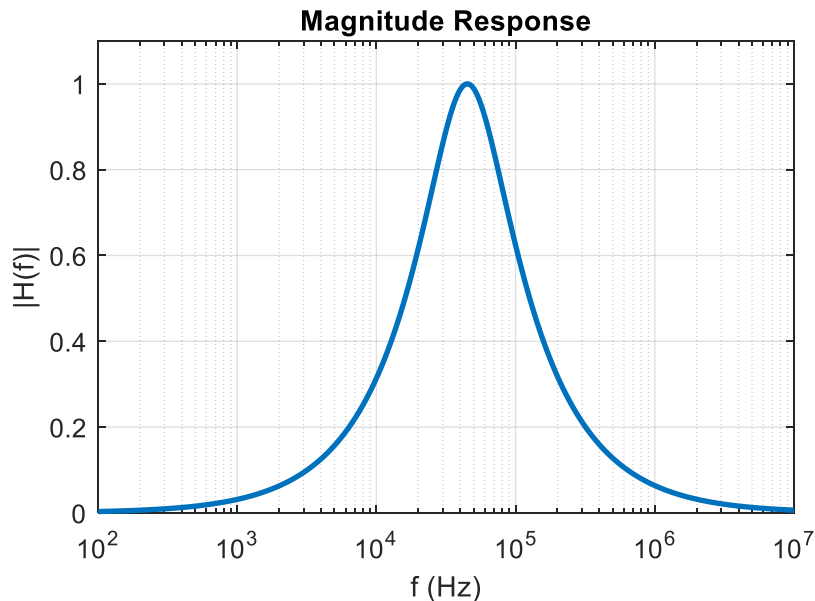
Magnitude Response - BPF



For the previous example, the magnitude response of the RLC circuit was

$$|H(\omega)| = \frac{\omega}{\sqrt{\left(200,000 - \frac{\omega^2}{400,000}\right)^2 + \omega^2}}.$$

- ✓ For low frequencies, $|H(\omega)| \approx 0 \rightarrow |Y(\omega)| \approx 0$,
- ✓ For high frequencies, $|H(\omega)| \approx 0 \rightarrow |Y(\omega)| \approx 0$.
- ✓ Only moderate frequencies will pass. $\rightarrow$ This is a band-pass filter.



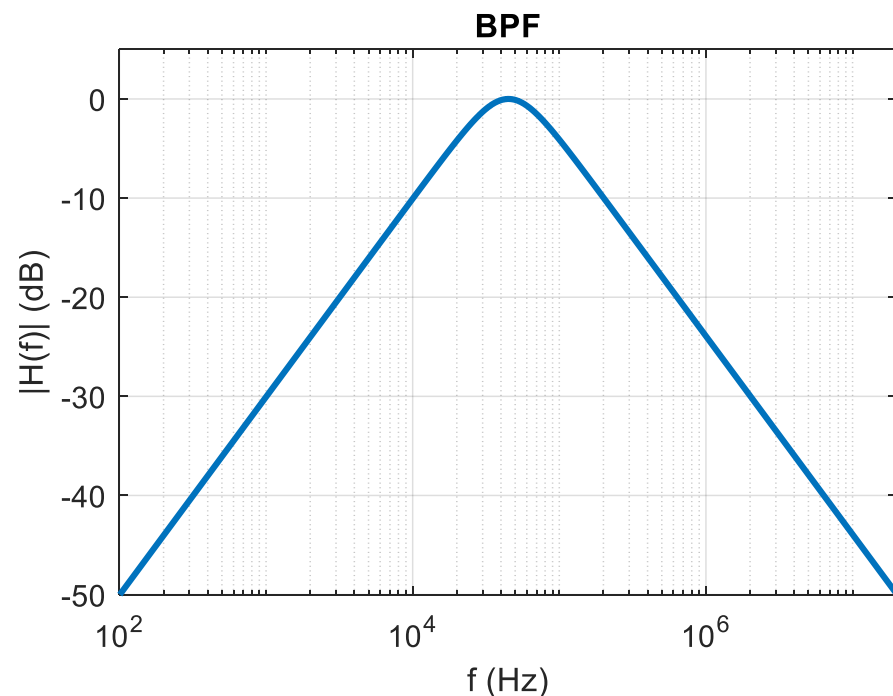
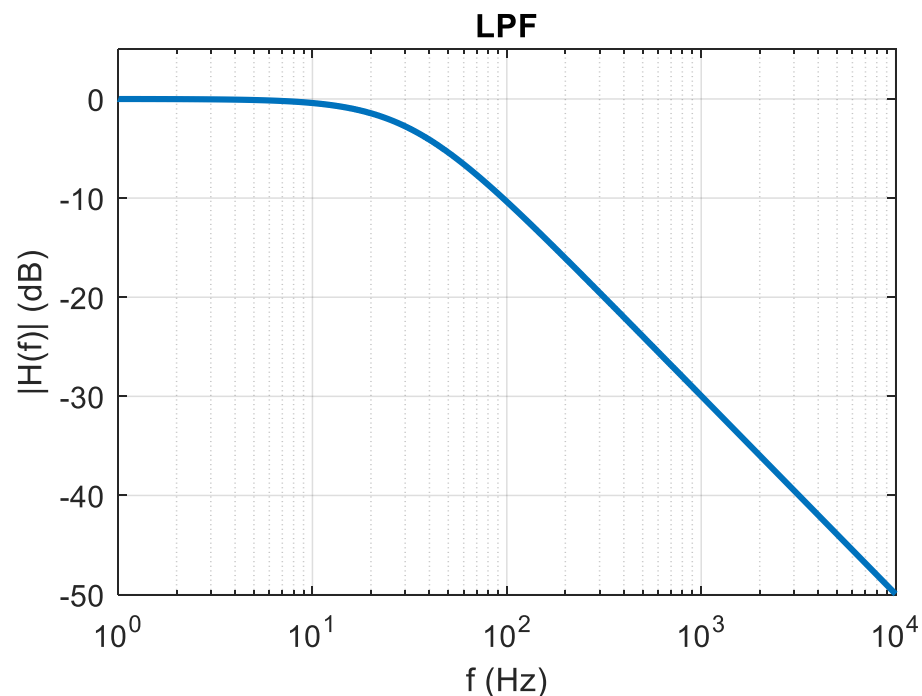
Decibels



It is common to plot the magnitude response on a log-log scale. This is often done by introducing the units of decibel (dB):

$$|H| \text{ in dB} = 20 \log_{10}(|H|) \quad \text{or equivalently} \quad |H|^2 \text{ in dB} = 10 \log_{10}(|H|^2)$$

For the two previous examples, the plots of the magnitude responses in dB are shown below. Note that using the dB format on a log-frequency scale, the plots are almost piecewise linear.



Decibels



To help give some insight to the decibel unit, the following table shows some conversions from absolute numbers to dB numbers.

$ H $	$ H ^2$	dB
1	1	0
$\sqrt{2}$	2	≈ 3
2	4	≈ 6
$\sqrt{5}$	5	≈ 7
$2\sqrt{2}$	8	≈ 9
$\sqrt{10}$	10	10
10	100	20
$10^{n/2}$	10^n	$10n$

$ H $	$ H ^2$	dB
1	1	0
$\sqrt{2}/2$	$1/2$	≈ -3
$1/2$	$1/4$	≈ -6
$\sqrt{5}/5$	$1/5$	≈ -7
$\sqrt{2}/4$	$1/8$	≈ -9
$\sqrt{10}/10$	$1/10$	-10
$1/10$	$1/100$	-20
$10^{-n/2}$	10^{-n}	$-10n$

Note: We can get from dB numbers back to magnitudes using:

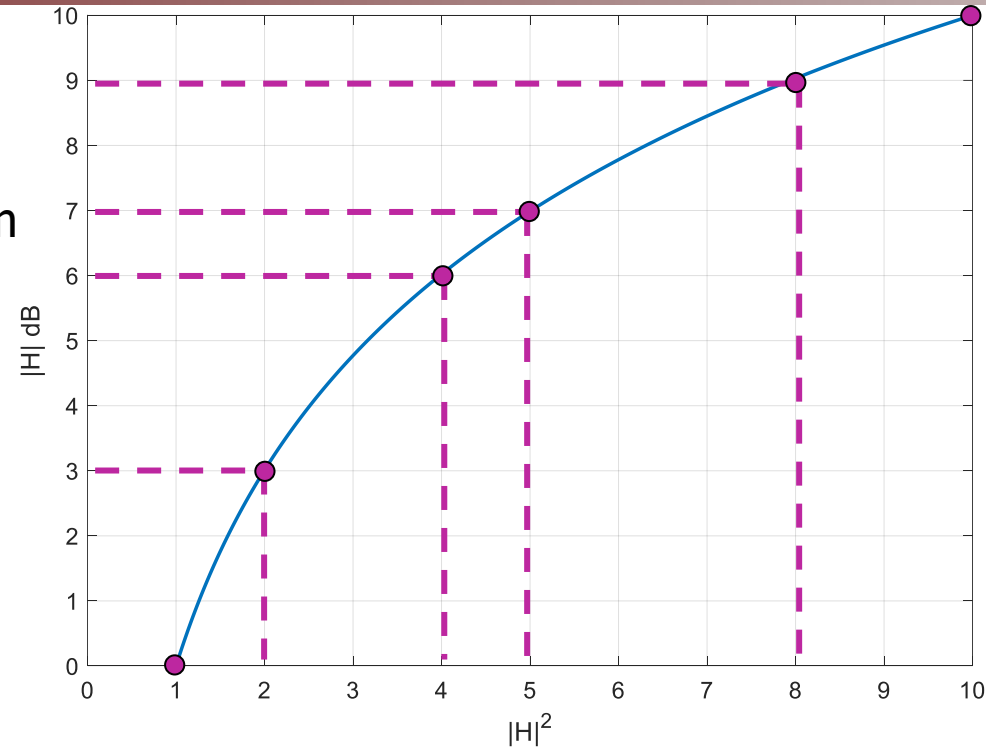
$$|H| = 10^{(|H|_{dB}/20)} \quad \text{or} \quad |H|^2 = 10^{(|H|_{dB}/10)}$$

Decibels



The following properties of decibels will help you to do dB calculations quickly in your head without having to need to pull out the calculator. These properties stem from log properties.

- If $|H|^2 \rightarrow x \text{ dB}$, then $2|H|^2 \rightarrow x + 3 \text{ dB}$,
- If $|H|^2 \rightarrow x \text{ dB}$, then $10|H|^2 \rightarrow x + 10 \text{ dB}$,
- If $|H|^2 \rightarrow x \text{ dB}$, then $\frac{|H|^2}{2} \rightarrow x - 3 \text{ dB}$,
- If $|H|^2 \rightarrow x \text{ dB}$, then $\frac{|H|^2}{10} \rightarrow x - 10 \text{ dB}$,



The decibel is a measure of the relative strength of two signals. When comparing signals in terms of power (or energy) the decibel is computed according to:

$$\text{dB} = 10 \log_{10}(P_1/P_2).$$

When comparing signals in terms of voltage (or current) the decibel is computed according to:

$$\text{dB} = 20 \log_{10}(V_1/V_2).$$

Since $P \propto V^2$, the dB measurement comes out the same regardless of whether we compare voltages/currents or power/energy.

Decibels



Example 13.3: Without using a calculator, convert each of the following numbers to decibels. Then check your answers with a calculator.

- a) $|H|^2 = 20,$
- b) $|H|^2 = 400,$
- c) $|H|^2 = 0.08,$
- d) $|H|^2 = 5 \times 10^{-6}.$

Example 13.4: Without using a calculator, convert each of the following dB numbers to magnitudes. Then check your answers with a calculator.

- a) $|H|^2 = 20\text{dB},$
- b) $|H|^2 = 33\text{dB},$
- c) $|H|^2 = -17\text{dB},$
- d) $|H|^2 = -36\text{dB}.$

Impulse Response



The inverse Fourier transform of the transfer function is called the *impulse response* of the circuit/system and is given by

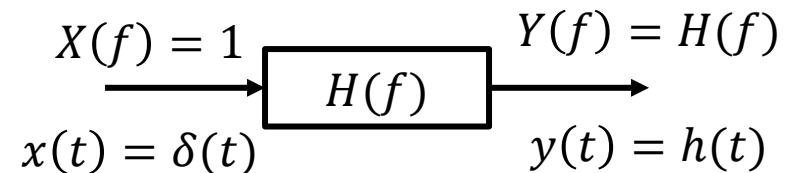
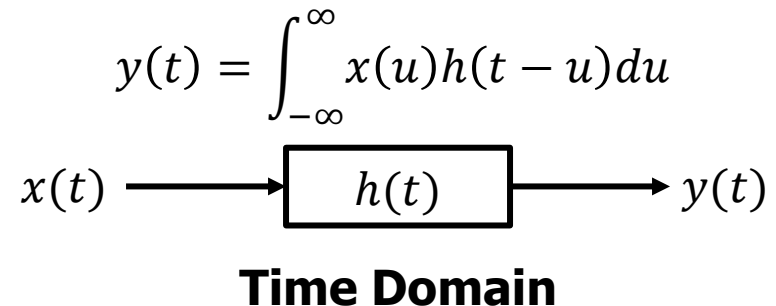
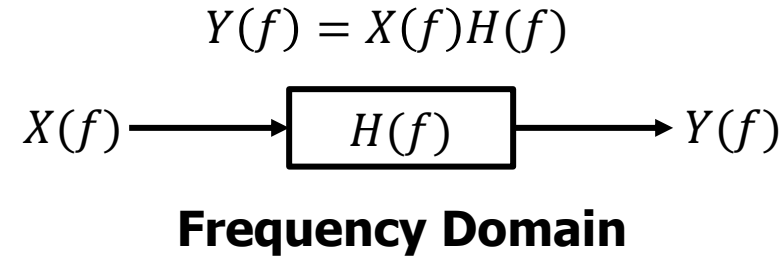
$$h(t) = \mathcal{F}^{-1}[H(f)] = \int_{-\infty}^{\infty} H(f) e^{j2\pi f t} df.$$

Recall that the transfer function relates the input and output in the frequency domain, $Y(f) = H(f) X(f)$.

Using the convolution property of Fourier transforms, this relationship in the time domain is

$$y(t) = \int_{-\infty}^{\infty} x(u) h(t - u) du = \int_{-\infty}^{\infty} x(t - u) h(u) du.$$

The impulse response gives the relationship between the input and the output of a system in the time domain through a convolution.



Impulse Response



Example 13.5: Find the impulse response of the RC low-pass filter that has a transfer function of

$$H(f) = \frac{1}{1+j2\pi fRC} .$$

Answer:

$$h(t) = \frac{1}{RC} \exp\left(-\frac{t}{RC}\right) u(t) .$$

Impulse Response



Example 13.6: Find the impulse response of the RLC band-pass filter that has a transfer function of

$$H(\omega) = \frac{j\omega}{8 - \omega^2 + 4j\omega},$$

Answer:

$$h(t) = e^{-2t}[\cos(2t) - \sin(2t)]u(t).$$